



231/1

BIOLOGY PAPER 1
MARKING SCHEME

1. These are genes located on the sex chromosomes and are transmitted together with those determine sex.

1 mk

2. (i) C – G – G – C – T – A – A – A – T – G – C – C **1 mk**

- (ii) C – G – G – C – U – A – A – A – U – G – C – C **1 mk**

3.

- The growing and developing regions such as shoots, leaves, flowers, fruits and roots
- Storage organs or tissues such as tubers, corns, bulbs, rhizomes and seeds.
- The secretors organs such as nectar glands in some insect pollinated plants such as bananas

any 3x1 = 3 mks

4. (i) Deoxygenated and oxygenated blood do not mix
(ii) Blood is at a higher pressure once the heart pumps it twice.

2x1 = 2 mks

5. (a)

- Number of limbs
- Presence and number of antennae
- Number of body parts

3x1 = 3 mks

(b)

i) Bryophyta

X – Seta

Z – Rhizoid rej Rhizoids.

6. A species is a subset of genus i.e. one genus contains several species. **1x1 = 1 mk**

7. Magnification = $\frac{\text{Length of drawing}}{\text{Length of real object}}$ ^{v1}

$$= \frac{3\text{cm}}{7\text{cm}} = 0.429 \text{ } ^{v1}$$

$$\times 0.43 \text{ } ^{v1}$$

1x3 = 3 mks

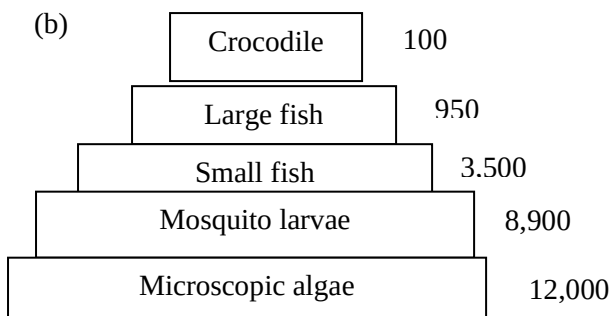
8. Because it involves movement of solvent (water) molecules from their region of high concentration to region of low concentration across a semi permeable membrane.

1x1 = 1mk

9. (a) Microscopic algae → mosquito larvae → small fish → large fish → crocodile

NB: mark as a whole**1x1 = 1 mk**

(b)

**1x1 = 1 mk**



- (c) i. Body size of the organism increase at each trophic level from the base as their numbers decrease.
 (ii). At each trophic level much of the energy obtained is lost in respiration thus fewer organisms can be supported at the succeeding level. **2 mks**
10. (a) Thin- To reduce distance for diffusion of gases.
 - To reduce distance for sunlight to reach the photosynthetic cells.
1x1 = 1 mks
 Broad – To provide large surface area for maximum light absorption.
1x1 = 1 mks
- (b) Pepsin – Breaks down proteins into peptides.
1x1 = 1 mks
 Renin – Digests protein caseinogens in milk to casein (curd)
1x1 = 1 mks
11. (i) Moist to dissolve the diffusing gases across the respiratory surface. **2 mks**
 (ii) Thin to reduce distance covered by diffusing gases i.e. for the gases to diffuse through short distance.
1mks
 (iii) Many palisade cells in a small area to enable them receive maximum sunlight.
1x1 = 1 mk
12. (a) A clinostat is a device which slowly rotates a plant to nullify the effect of unidirectional stimulus.
1x1 = mk
 (b) Horizontal direction as in diagram.
1mks
 (c) The clinostat will nullify the effect of gravity on the seedling hence the seedling continues to grow horizontally.
1x1 = 1 mk
13. This is because during humid day, there is low rate of sweating; since less water is lost from the body surface, leading to less heat loss through sweat hence body temperature tend to rise slightly.
1x2 = 2 mks
14. High concentration of rods in the retina of their eyes.
 - There is more rods than cones in the retina.
1x2 = 2 mks
15. (i) Helps in the transport of proteins. **1 mk**
 (ii) Manufacture of ribosomes **1 mk**
16.
 - Stomata;
 - Lenticels of woody plants
 - Cuticles
3x1 = 3 mks
17. (a) Pitfall trap
 (b) For catching crawling animals.
18.
 - Distortion of parts of fossils during sedimentation hence can give wrong impression of the structure;
 - There was several missing links of fossils records as some parts or whole organism decomposed, some scavenged upon and conditions may not be conducive for fossilization (O.W.T.T.E).
 - Destruction of fossils by geological activities like earthquakes, faulting and mass movement.
3x1 = 3 mks
19. Enzymes cofactors are non-proteinous substances which activate enzymes; while co-enzymes are organic non-protein molecules that work in association with particular enzymes. (Mark as a whole)
2 mks
20.
 - Cells have adjusted to the new environment
 - Food and other factors are not limiting hence no competition for resources.



- Rate of cell increase is higher than cell death.
- There is an increase in the number of cells dividing

First two **1x2 = 2 mks**

21.

- Latin language was widely spoke and used by scientists during his time;
- Local names used previously could not be understood by everyone thus Latin language enhanced scientific communication worldwide.

1x2 = 2 mks

22.

- Excretion of waste products from the body cells.
- Absorption of digested food from alimentary canal of animals in the blood stream.
- Absorption of some minerals salt from soil by plant roots.
- Accumulation of substances into the body to offset osmotic imbalance in arid and saline environment.
- Reabsorption of sugars and some salts by the kidney.

3x1 = 3 mks

23.

- Oxygen concentration
- Presence or absence of hormones
- Substrate concentration
- Surface area to volume ratio/body size of an organism.

3x1 = 3 mks

24. (a) Mature human ovum

1mk

(b) X – follicle cell

Y – viteline membrane

1mk

W – Plasma membrane

1mk

25. (a) Femurs

(b) Q – neck

R – Shaft

(c) Structure - patella

Joints – hinge joint

26.

- Act as a cushion that absorbs shock thus reducing friction.
- Allows for a certain degree of movement between the vertebrae in the vertebral column.

2x1 = 2 mks

27. Some malaria plasmodium developed resistance; to chloroquine drug; through mutation; those resistant individuals transmit the characteristic to their offspring through reproduction thus establishing a new population of resistant forms.

4 mks

28.

- It is long to provide large surface area for absorption.
- It is numerous to bring digested food into close contact with walls of the ileum for easier absorption.
- Highly coiled to slow down movement of food, allowing more time for absorption.
- Higher surface has large number of villi and micro-villi which increase the surface area for absorption of end products of digestion.
- Presence of thin layer of cells through which digested food diffuses.
- Presence of tense network of blood capillaries in villi into which nutrients are absorbed.
- Presence of lacteals in the villi for absorption of fatty acids and glycerol.

4x1 = 4 mks

29.

- Independent assortment
- Crossing over

2x1 = 2 mks